THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS TWO-CHAPTER TEST SET D

Chapters 1 & 2: **Patterns in Mathematics, Lines and Angles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The next term in 5, 10, 20, 40, … is: 1

(a) 80 **(b)** 60 **(c)** 70 **(d)** 100

2. Hexagonal numbers 1, 7, 19, 37, … are formed by: 1

(a) dots arranged in hexagons around a centre
(b) dots in a line
(c) dots in a triangle
(d) dots in a square grid

3. The 6th triangular number is: 1

(a) 19 **(b)** 27 **(c)** 21 **(d)** 22

4. The sum of the first 7 odd numbers is: 1

(a) 56 **(b)** 42 **(c)** 49 **(d)** 14

5. The sum of the first 25 odd numbers is: 1

(a) 625 **(b)** 50 **(c)** 600 **(d)** 650

6. The angle between North and East is: 1

(a) 270° **(b)** 180° **(c)** 45° **(d)** 90°

7. The number of degrees in half a turn is: 1

(a) 180 **(b)** 90 **(c)** 270 **(d)** 360

8. An angle of 45° is: 1

(a) straight **(b)** obtuse **(c)** acute **(d)** right

9. An angle of 300° is: 1

(a) acute **(b)** obtuse **(c)** reflex **(d)** straight

10. An angle of 181° is: 1

(a) obtuse **(b)** acute **(c)** right **(d)** reflex

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Is 100 a square number? Give a reason. 2

12. Add two consecutive triangular numbers 15 and 21. What do you get? 2

13. How do you read angles from a protractor correctly? 2

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14. Name two places where you see angles in everyday life. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain how the Koch snowflake is built by repeating a rule. 3

16. Estimate and then measure: how would you check the measure of the angle formed by a laptop screen? 3

17. Find the 15th odd number and the sum of the first 15 odd numbers. 3

OR

Observe 1×1=1, 11×11=121, 111×111=12321. Predict 1111×1111. 3

18. Name the types of angle formed by the hands of a clock at 1:00, 3:00 and 5:00. 3

OR

Why is the measure of an angle independent of the length of its arms? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Write a story problem that leads to a number pattern. 5
(ii) Solve it for 6 terms.
(iii) State the rule.

OR

(i) Explain the idea of ‘mathematics is the study of patterns’. 5
(ii) Give three examples from nature or daily life.
(iii) Why is it important?

20. (i) Describe an experiment using paper folding to get a right angle. 5
(ii) Use it to find an angle of 45°.

OR

(i) Why are tracks of a railway drawn parallel? 5
(ii) Give two examples of intersecting lines in surroundings.

END OF QUESTION PAPER — Total: 40 Marks

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